

Splitting the Difference

Team Members: _____

Your aunt and uncle are getting divorced. Rather than argue over who gets what, they have agreed to settle it with a coin. For each item on the household inventory they will flip once: heads to your aunt, tails to your uncle. Nobody chooses, nobody negotiates, nobody can accuse the other of gaming it. Each item is a fair, independent flip. They are pleased with themselves. Your job is to find out whether they should be.

Part 1. Predict First

Answer both of these *before* you touch the coin. Write down a number, not a feeling.

1.1 The inventory below is appraised at \$120,000. How much do you expect your aunt to end up with? _____

1.2 How far from an even split do you expect the actual settlement to land? Circle one.

under \$5,000

\$5,000–15,000

\$15,000–30,000

over \$30,000

Part 2. The Settlement

One of you is the AUNT and takes heads. One of you is the UNCLE and takes tails. Flip once for each of the twelve items and mark the winner with **A** or **U**. Then do the whole thing a second time, from scratch — a second settlement, in a parallel universe where the coin fell differently.

	Item	Appraised value	Round 1	Round 2
1	The lake cabin	\$60,000		
2	Pickup truck	\$14,000		
3	Sailboat	\$11,000		
4	Grandmother's diamond ring	\$8,500		
5	Baby grand piano	\$7,000		
6	Comic book collection	\$5,500		
7	Dining room set	\$4,500		
8	Snowblower and power tools	\$3,500		
9	Espresso machine	\$2,500		
10	Mountain bikes (pair)	\$2,000		
11	Vinyl record collection	\$1,500		
12	Bruno, the dog	\$0		
	Total	\$120,000		

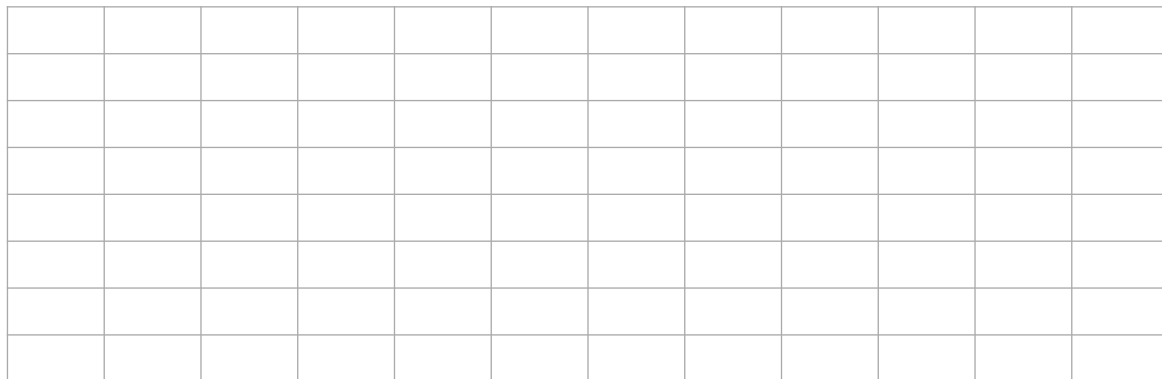
2.1 Round 1 — Aunt: _____ **Uncle:** _____ **Gap:** _____

Round 2 — Aunt: _____ **Uncle:** _____ **Gap:** _____

Report both of your aunt's totals to the board.

Part 3. The Class Distribution

Every settlement in the room goes on the board as one mark in the bin containing the aunt's total. Copy the finished class histogram here.



0–10 10–20 20–30 30–40 40–50 50–60 60–70 70–80 80–90 90–100 100–110 110–120

Aunt's share, thousands of dollars

3.1 Where is the center of the class distribution? _____

An even split would be \$60,000. How close is the center to that?

3.2 Settlements within \$10,000 of even: _____ out of _____ That is _____ percent.

3.3 The histogram does not have a single hump in the middle. Describe its shape, then explain what feature of the inventory produces that shape.

3.4 Largest gap between the two parties seen anywhere in the room: _____

3.5 The average settlement is \$60,000, and \$60,000 sits in one of the emptiest parts of the histogram. Explain how a number can be the average of a quantity and still be an outcome you almost never see.

Part 4. Discussion

4.1 Each flip is fair and the procedure gives each party an expected \$60,000. Your uncle says that makes the settlement fair. Your aunt, holding the record collection and the espresso machine, disagrees. Both of them are using the word correctly. What are the two different things they mean by it?

4.2 Bruno is appraised at \$0, and the coin treats him accordingly. Name one other item on the list that is probably worth very different amounts to the two of them, and say what the appraisal column is failing to capture.

4.3 Suppose the cabin were sold first and replaced on the inventory by four items worth \$15,000 each. The total is unchanged and each party's expected share is still \$60,000. What happens to the histogram? Would you call the new procedure fairer, and in which of the two senses from 4.1?

4.4 Propose a settlement procedure you would actually recommend to them. What does your procedure require from your aunt and uncle that the coin does not require?

4.5 Every flip was fair, every flip was independent, and the expected shares were equal. So where did the unfairness come from?

